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THE INTERNATIONAL UNIVERSITY**

SCHOOL OF COMPUTER SCIENCE AND ENGINEERING



**WEB APPLICATION DEVELOPMENT LAB
IT093IU - Group 02**

PROJECT REPORT

Topic: Food Recipe Website

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CHAPTER 1: INTRODUCTION

1.1 Abstract:

This web application development project involves building an online Food Recipe Website. The project is developed using Java for the back-end and MySQL for database management, with GitHub support to track the project's progress and enable collaboration among team members. The project also emphasizes storing information about. The database structure is carefully designed to maintain system scalability and efficiency, supporting smooth operation across all processes. Furthermore, the design process defines data entities, relationships, attributes, and constraints logically to maintain data integrity. The user interface is designed to be intuitive, enabling users to check for food categories, search for specific food recipes, and add or edit recipes, while administrators can manage users, check statistics and generate reports.

1.2 Project Overview:

A Food Recipe Website allows users to register, log in and add recipes with different access levels (user, administrator). Users can search for food recipes, select a desired recipe. Administrators can manage the recipes data. All data including user information, recipe details will be stored and managed using MySQL. The admins can delete it. The database is supported with Entity Relationship Diagrams (ERDs) and schema diagrams to provide a clear visualization of entities and relationships, ensuring efficient SQL queries and reliable data operations.

1.3 System Overview

- **User Interface (UI):**
 - Designed to be user-friendly and easy to navigate without being too complex.
- **Product Display:**
 - Our homepage of the website allows users to view their desired recipe in many different categories and tags.
- **User Registration and Login:**
 - Users can register a new account then login that registered account to be able to add a new recipe on.
- **View Recipe:**
 - When users click on a recipe, they can see the information of different ingredients and tips, timer of each stage of cooking.
- **Add Recipe:**
 - Users are able to add a new recipe, include detailed ingredient information of how to cook, timer of each stage of cooking.

1.4 Tool & Environment:

This project utilized the following tools and environment:

HTML/CSS: Website main framework and format the HTML webpage in a unique-looking way.

JS: for backend development to handle logic and recipe details.

Draw.io: Design ER Diagram and Relational Schema.

1.5 Installation & Setup:

Clone the repository: `git clone `HeomapTV123/Recipes.github.io``

To set up the project, run “run npm dotenv” in the backend folder to access the session token and port to host the server. The configuration can be changed as desired, but this project is built with “Port: 5000”.

To get the website running, open a browser and put

“localhost:5000/html/main.html” to access the main homepage.

CHAPTER 2: METHODOLOGY

2.1 Requirement Analysis:

This project is a recipe management platform that allows users to create, manage, categorize, and interact with cooking recipes. The system supports user accounts, recipe publishing, ingredients, step-by-step instructions, tagging, categorization, comments, ratings, and favorites.

2.2 Approach Analysis:

2.2.1 Review Materials:

- NodeJS
- HTML, CSS, JavaScript, mySQL

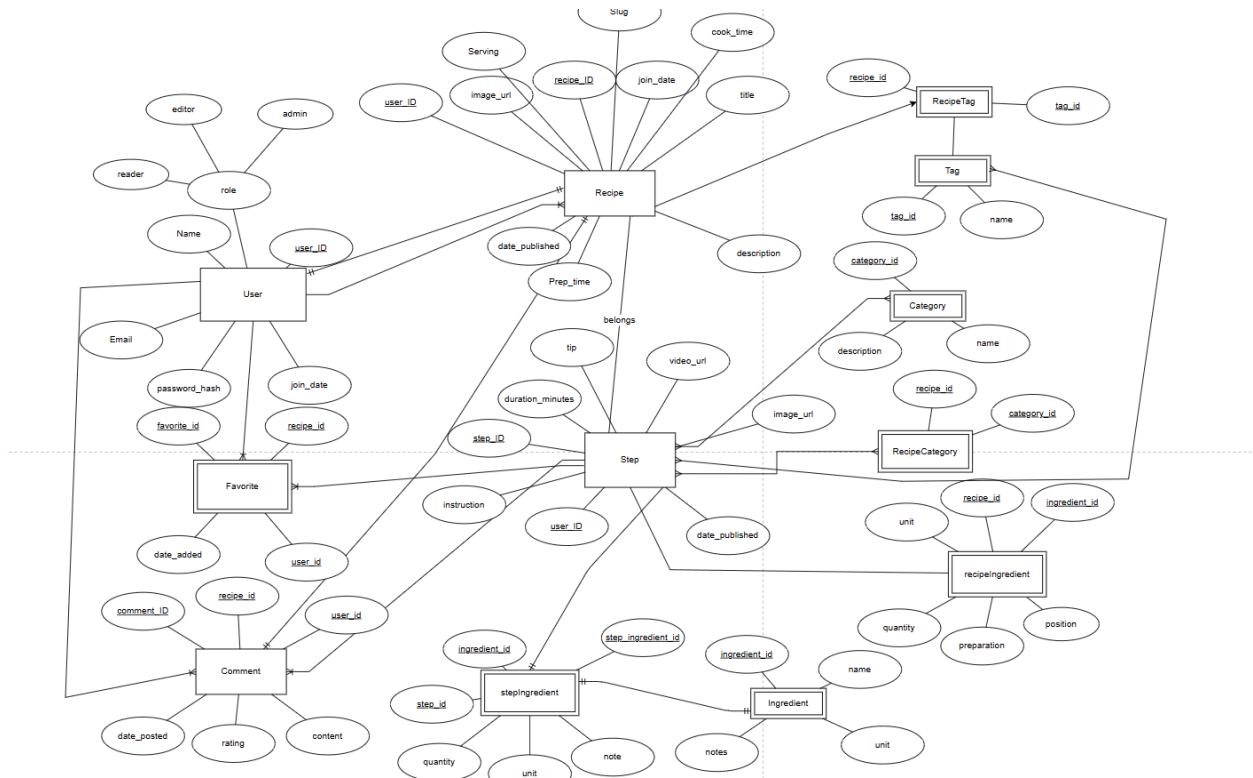
2.2.2 Research Approach:

- Learning Syntax of required programming languages (HTML, CSS, JavaScript, MySQL, NodeJS)

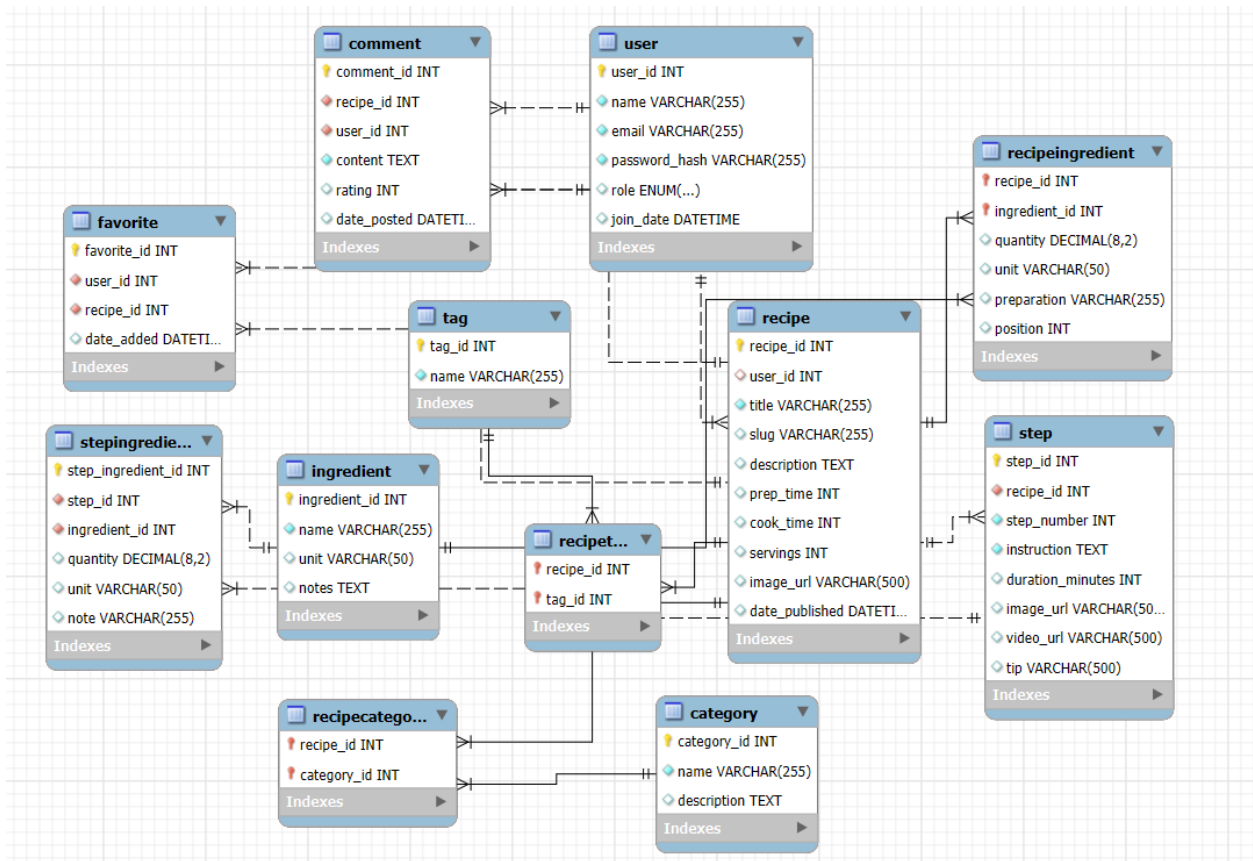
2.2.3 System Analysis:

- Strategic Choices in System Development:
 - + Front-end: HTML, CSS, JavaScript
 - + Back-end: NodeJS
 - + Database: MySQL

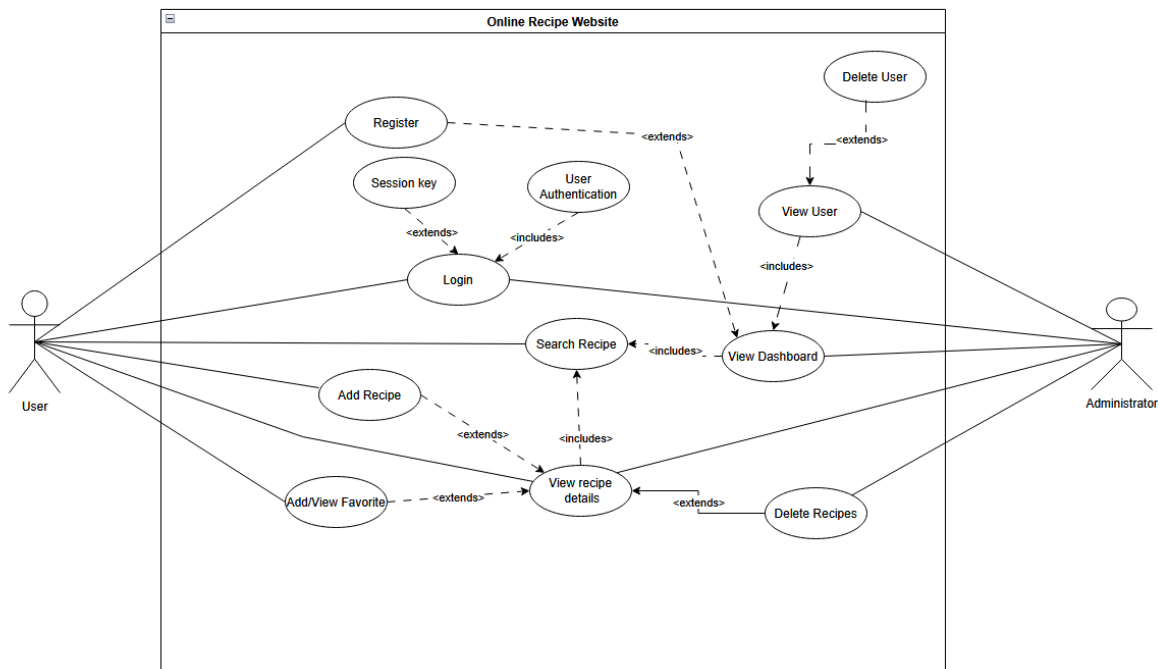
2.3 ERD:



2.4 Schematic:



2.5 Use Case Diagram:



2.6 Use Cases:

Use case 1:

Name case: Log in to the recipe website

Identifier: UC1

Input:

1. Username
2. Password

Output:

1. The homepage if the credentials are valid
2. The login page otherwise

Basic Course

Actor: User (Guest), Admin	System
1. Open the login page	1.1 Display the login page
2. Type in the username, password	
3. Submit	3.1 Check for the user's info 3.2 Redirect to the homepage if success 3.3 Else return to login page

Precondition

1. User must have a registered account

Post condition

1. None

User story: As a visitor, I want to log in to the recipe website so that I can look up recipes and save the recipes that I want to make.

Use case 2:

Name case: Register an account on the food recipe website

Identifier: UC2

Input:

1. Email

2. Username
3. Password

Output:

1. The homepage if success
2. The signup page if fails

Basic Course

Actor: User (Guest)	System
1. Open the sign up page	1.1 Display the login page
2. Type in the email, username, and password	
3. Submit	3.1 Check if user's info is valid 3.2 Redirect to main page if success 3.3 Sign up page if fail

Precondition

1. None

Post condition

1. User account added into database

User story: As a visitor, I want to register an account so that I can save recipes, submit my own recipes, and personalize my experience on the website.

Use case 3:

Name case: Search a recipe by name, or tags

Identifier: UC3

Input:

1. Search keyword or tag

Output:

1. The result page with all of the recipes containing the keyword or tags

Basic Course

Actor: User (Guest)	System
1. Open the homepage	1.1 Display the homepage
2. Click on the search bar	
3. Type in the keyword or tags to search for the recipe	
4. Submit	4.1 Select the recipes that match with the search condition from the database 4.2 Display the result page and the search result

Precondition

1. None

Post condition

1. Result page is displayed

User story: As a visitor, I want to search for a recipe with the search bar so I can quickly find meals that match my ingredients, preferences, and dietary needs without wasting time browsing unrelated recipes.

Use case 4:

Name case: Save/Remove a recipe into/from the user's favorites

Identifier: UC4

Input:

1. None

Output:

2. Display a message that user has saved the recipe successfully

Basic Course

Actor: User	System
1. Click the hollow heart icon on the recipe card	1.1 If the user is logged in, add recipe to the saves page 1. 2 Else, then display message "User must logged in to save a recipe"
2. Click the filled heart icon on the recipe card	2.2 Remove the recipe from the saves page

Precondition

1. User is logged in

Post condition

1. Recipe is added/removed to/from the saves page
2. Database updated

User story: As a user, I want to add/remove a recipe to/from my saves page because I want to manage a personalized collection of recipes that I want to cook or visit later.

Use case 5:

Name case: View favorites

Identifier: UC5

Input:

1. Click on the star icon next to the user profile

Output:

1. Display the saves page with all the favorite recipes

Basic Course

Actor: User (Guest)	System
1. Click on the star icon	1.1 Display the saves page

Precondition

1. User is logged in

Post condition

1. The saves page is displayed

User story: As a user, I want to click on my favorites because I want to access all of the meals I want to prepare and cook later.

Use case 6:

Name case: View recipe's details

Identifier: UC6

Input:

1. Click on the recipe card

Output:

1. Display the recipe page with all the instructions

Basic Course

Actor: User (Guest)	System
1. Click on the recipe card	1.1 Display the recipe page

Precondition

1. None

Post condition

1. The recipe page is displayed

User story: As a user, I want to click on a recipe card so that I can view the ingredient list and step-by-step instructions so that I can prepare and cook the dish accurately.

Use case 7:

Name case: Display recipes by category

Identifier: UC7

Input:

1. Click on the category navigation bar

Output:

1. Display all the recipes that match the category

Basic Course

Actor: User (Guest)	System
1. Click on a category navigation button	1.1 Display the recipe page

Precondition

1. None

Post condition

1. Display recipes by clicked category

User story: As a user, I want to find a recipe by category so that I can quickly browse and select meals that fit a specific type, cuisine, or occasion

Use case 8:

Name case: Add a recipe

Identifier: UC8

Input:

1. Click on the Add Recipe Button
2. name, category, preparation time, cooking time, tags (optional), ingredients, quantity, unit, step instructions, image

Output:

1. Display message “Recipe added successfully.”
2. Redirect to homepage

Basic Course

Actor: User	System
1. Click on the Add Recipe button	1.1 Redirect to add recipe form page
2. Type in the recipe details name, category, prep and cooking time, tags, ingredients, quantity, unit, step instructions, duration, notes, upload image Delete an ingredient or a step instruction	2.1 Fetch all categories for the drop down list 2.2 Add/Delete an ingredient with quantity and unit input 2.3 Add/Delete an step with instructions, duration, and notes
3. Submit	3.1 Check if required fields are filled 3.2 Display message “Recipe added successfully.” and redirect to homepage if success 3.3 Return to add recipe page

	if fails.
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Precondition

1. User is logged in

Post condition

1. Recipe is added to the database

User story: As a user, I want to find a recipe by category so that I can quickly browse and select meals that fit a specific type, cuisine, or occasion

Use case 9:

Name case: Manage user account

Identifier: UC9

Input:

1. Click on the Manage User button

Output:

1. Display all of user's accounts

Basic Course

Actor: Admin	System
Log in with admin account	1.1 Verify if account has admin role 1.2 Redirect to admin

	dashboard if success 1.3 Return to login page if fails
Click on the Manage User button	2.1 Display all of the user's accounts from the database

Precondition

1. User is logged in
2. User has admin role

Post condition

1. Admin dashboard is displayed
2. All of the user's accounts are displayed

Admin story: As an admin, I want to manage all of my user's accounts so that I can verify my user's information

Use case 10:

Name case: Manage recipes

Identifier: UC10

Input:

1. Click on the Manage Recipe button

Output:

1. Display all of the recipes in the database

Basic Course

Actor: Admin	System
Log in with admin account	1.1 Verify if account has admin role 1.2 Redirect to admin dashboard if success 1.3 Return to login page if fails
Click on the Manage Recipe button	2.1 Display all of the recipes from the database

Precondition

1. User is logged in
2. User has admin role

Post condition

1. Admin dashboard is displayed
2. All of the recipes from the database are displayed

Admin story: As an admin, I want to manage all recipes so that I can moderate content and maintain quality.

Use case 11:

Name case: Delete a user account

Identifier: UC11

Input:

1. Click on the Delete User button

Output:

1. Alert message “User deleted successfully.”
2. Refresh the Admin Dashboard page

Basic Course

Actor: Admin	System
Log in with admin account	1.1 Verify if account has admin role 1.2 Redirect to admin dashboard if success 1.3 Return to login page if fails
Click on the Delete User button	2.1 Ask for admin’s confirmation “Are you sure you want to delete this user?” 2.2 Delete the user account from the database if yes 2.3 If no, then return to admin dashboard page

Precondition

1. User is logged in
2. User has admin role

Post condition

1. User account is deleted from database
2. Alert message “User deleted successfully.”

Admin story: As an admin, I want to delete a user account so that I can ban users violating the community rules, remove inactive or abandoned accounts

Use case 12:

Name case: Delete a recipe

Identifier: UC12

Input:

1. Click on the Delete Recipe button

Output:

1. Alert message “Recipe deleted successfully.”
2. Refresh the Admin Dashboard page

Basic Course

Actor: Admin	System
Log in with admin account	1.1 Verify if account has admin role 1.2 Redirect to admin dashboard if success 1.3 Return to login page if fails
Click on the Delete Recipe button	2.1 Ask for admin’s confirmation “Are you sure you want to delete this recipe?” 2.2 Delete the recipe from the database if yes 2.3 If no, then return to admin dashboard page

Precondition

1. User is logged in

2. User has admin role

Post condition

1. Recipe is deleted from database
2. Alert message “Recipe deleted successfully.”

Admin story: As an admin, I want to delete a recipe so that I can remove inappropriate, or offensive content, prevent recipe duplications and low-quality recipes.

2.7 Demonstration:

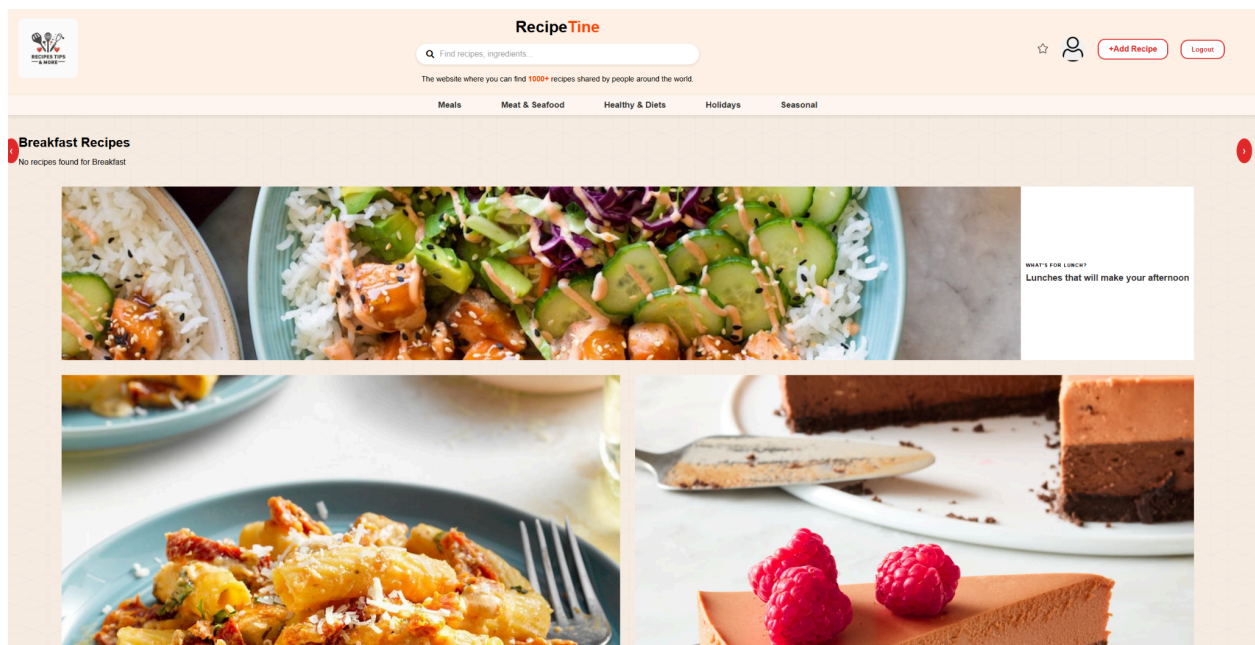


Figure 4.1: Homepage

Add Your Own Recipe!

Recipe Name

Category

Prep Time (in minutes)

Cook Time (in minutes)

Tags

Egg ✕
Breakfast ✕
Yummy ✕
Enter tags (e.g chick)

Unit Types

teaspoon = tsp.
tablespoon = tbsp.
cup = c.
quart = qt.
ounce = oz.
pint = pt.
gallon = gal.
inch = in.
pound = lb.
fahrenheit = F.

milliter = ml.
liter = L.
milligram = mg.
gram = g.
kilogram = kg.
millimeter = mm.
centimeter = cm.
meter = m.
celsius = C.

List of Ingredients

Ingredient	Quantity	Unit	
Salt	20	mg	Delete
Ingredient	Quantity	Unit	
Egg			Delete
Ingredient	Quantity	Unit	
Pepper	20	mg	Delete

+ Add Ingredient

Figure 4.2: Add recipe

Fried Rice

Fried Rice

Ingredients

✓ 2.00 cups of Rice

✓ 2.00 pcs of Eggs

✓ 2.00 tbsp of Soy Sauce

✓ 2.00 cloves of Garlic

✓ 2.00 stalks of Green Onions

✓ 0.50 cups of Carrots

Steps

Step 1
Heat oil in a pan and sauté garlic until fragrant.
Duration: 2 minutes

Step 2
Add carrots and stir-fry for 3 minutes.
Duration: 3 minutes

Step 3
Push vegetables aside and scramble eggs in the same pan.
Duration: 2 minutes

Step 4
Add rice, soy sauce, and green onions. Mix well and fry for 5 minutes.
Duration: 5 minutes

Figure 4.3: Recipe View with Steps

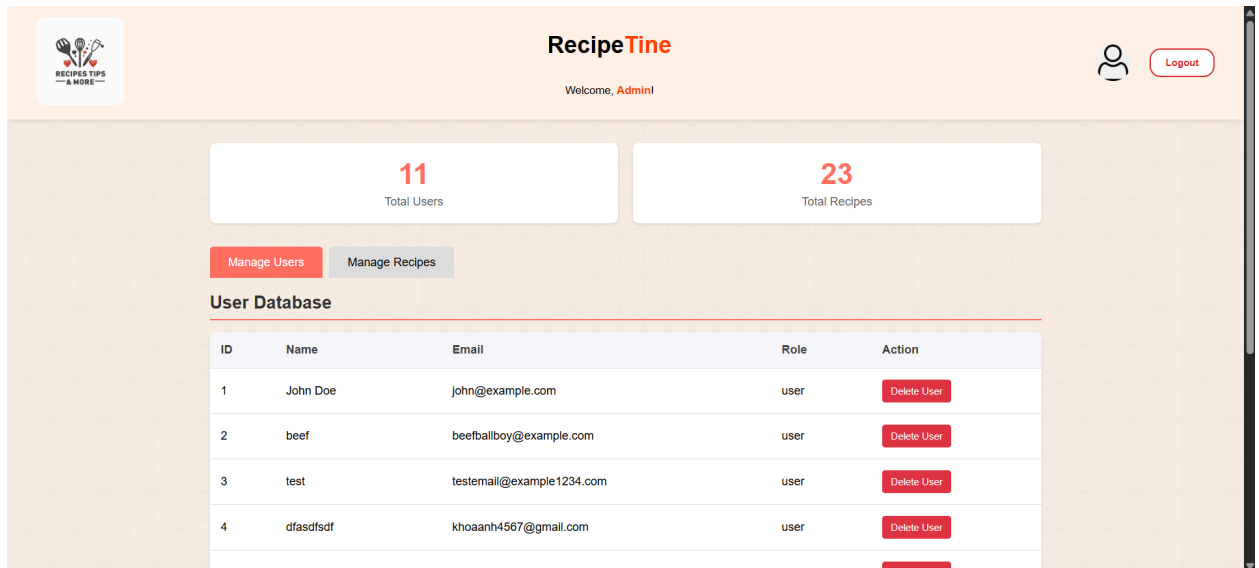


Figure 4.4: Admin Dashboard with delete user and recipe

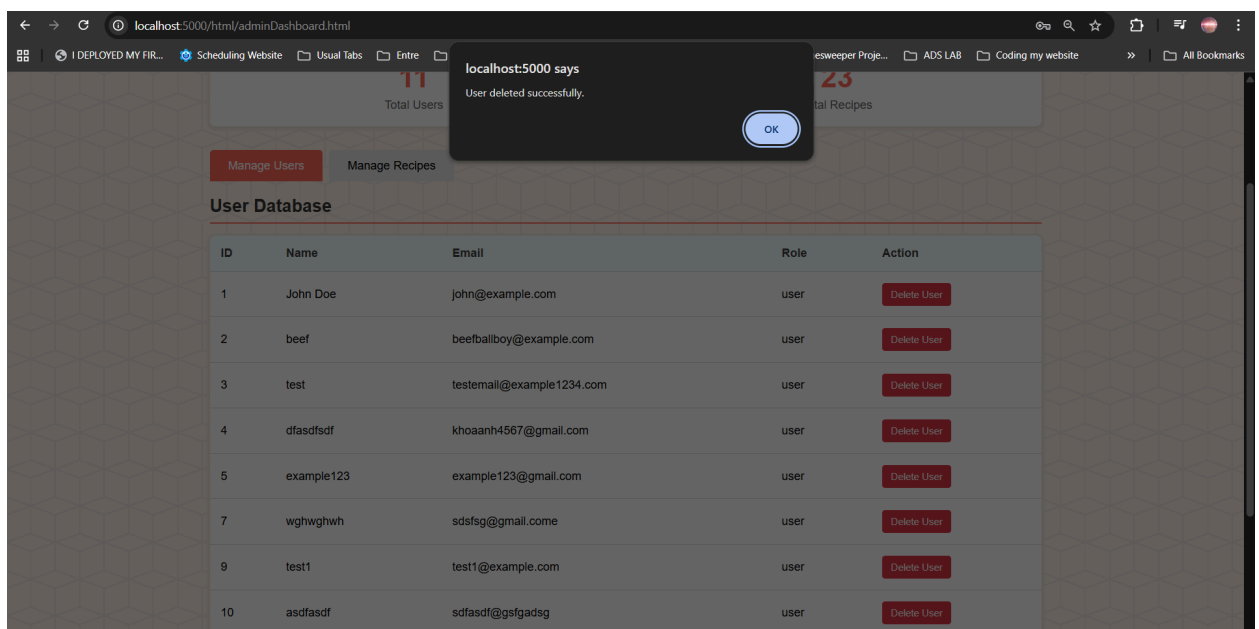


Figure 4.5: Delete user successfully

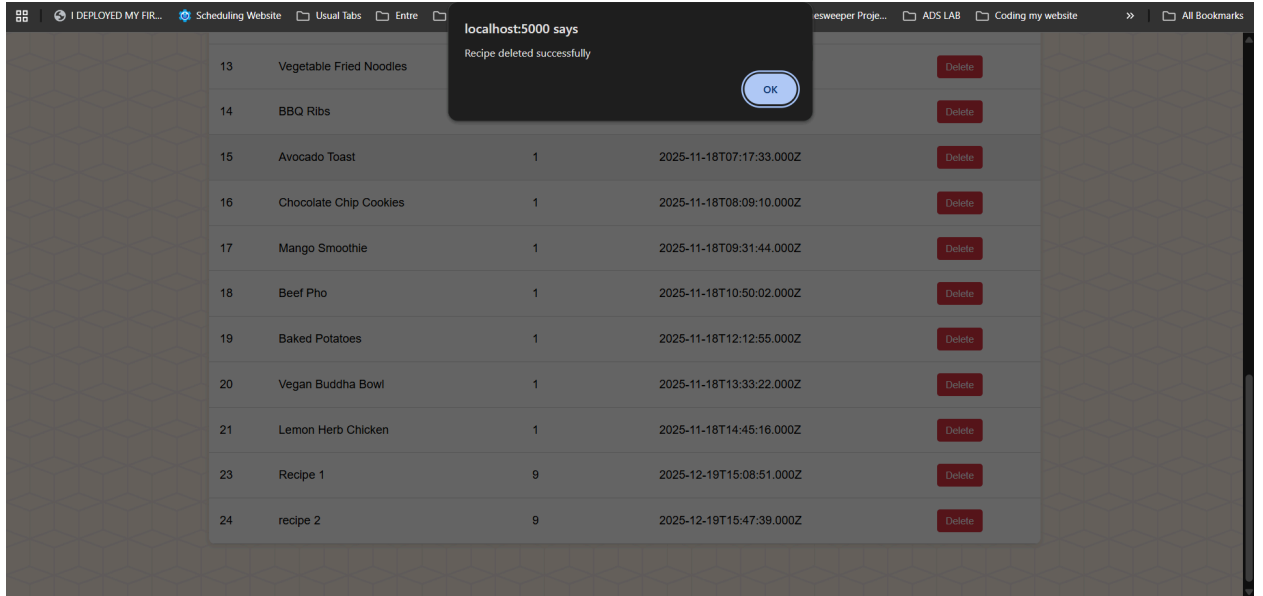


Figure 4.6: Delete recipe successfully

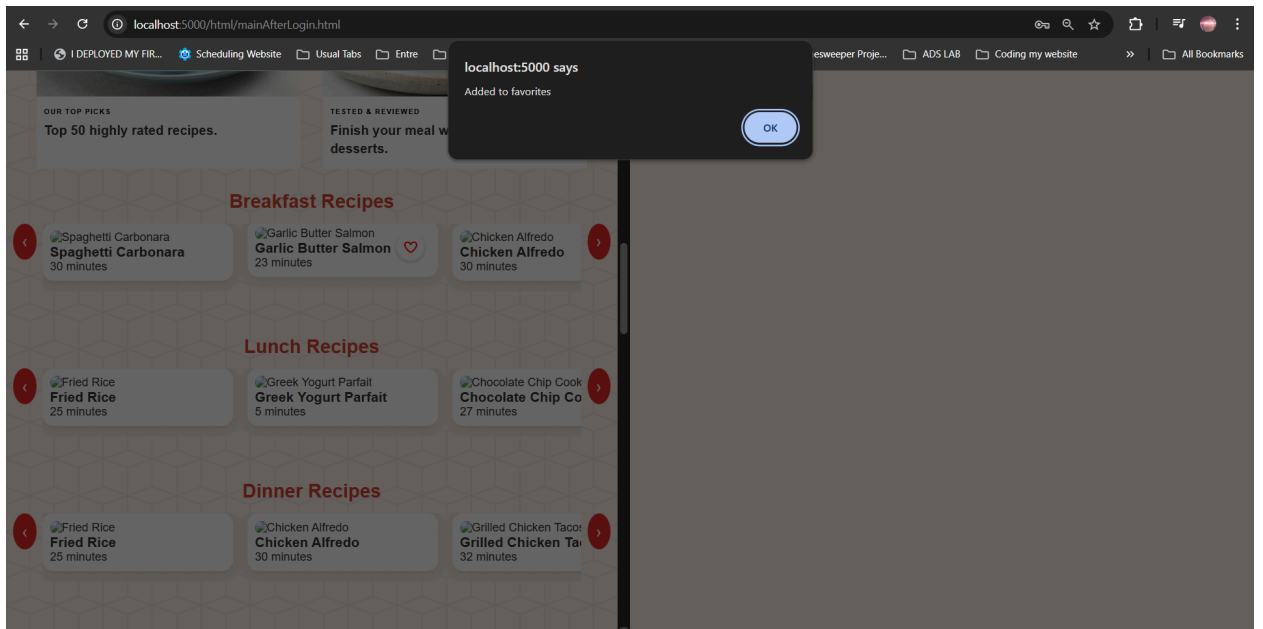


Figure 4.7: Add a recipe to favorites

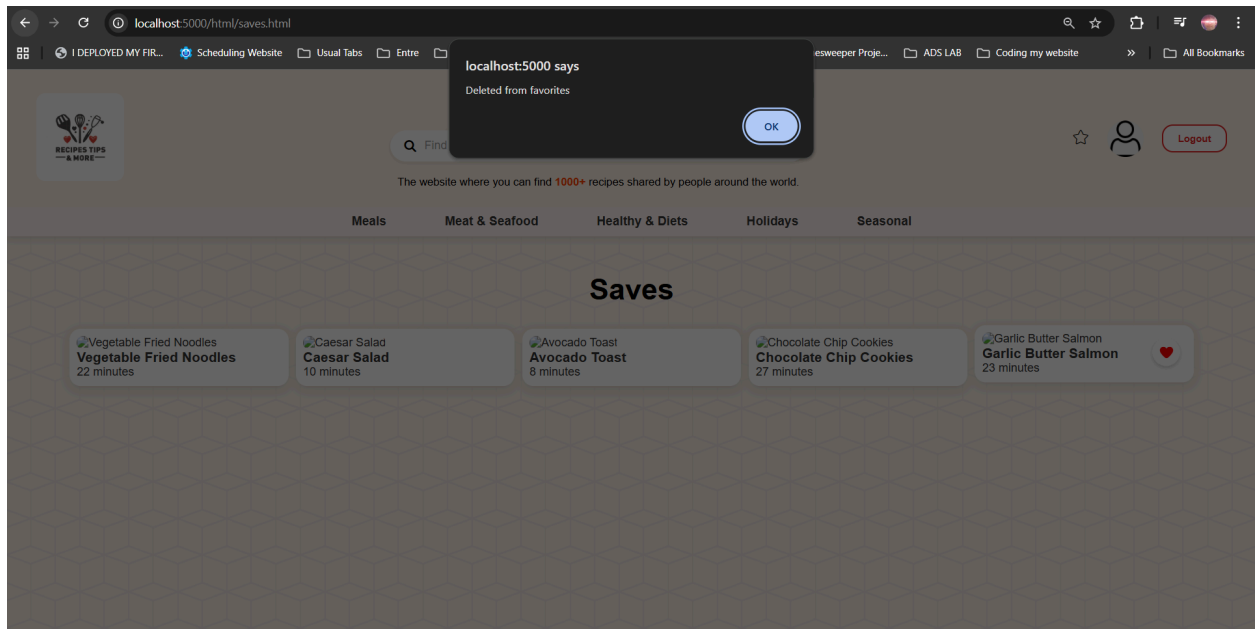


Figure 4.8: Unfavorite a recipe

CHAPTER 3: CONCLUSION

3.1 Conclusion

Key Findings and Achievements:

- + **Main Findings:** We successfully created a website for the online recipe that allows users to see recipes of many categories. Users can register, log in, rate recipes, comment on them.
- + **Achievements:** Implementing a friendly interactive interface for users to navigate, choose food recipe tags, users are able to add their own recipe and tag them.

Reflection on Objectives:

- + **Overall Objective Evaluation:** The project's main goals were all met. The website successfully displays services and costs, enables user registration and login, allows users to favorite desired recipe and admins to be able to delete recipes, and supports recipe information and administration management.
- + **Planned Features:** During development we got into some troubles with testing and there were many unplanned features such as proper edit recipe comment, average rating, video URL. We believe that in the future these features will be implemented in one way or another
- + **Technical Problems in Testing:** When testing the websites, we got into some troubles with a bugged search function where it displayed untagged and uncategorized food, then favorite feature did not work as intended and had to be clicked twice to undo. Lastly, the images which were not displayed properly on the main homepage.

3.2 Concluding Thoughts

The Food Recipe Website project provided an excellent opportunity to apply database management principles and web app development skills learned in the course. The project, developed collaboratively by the team, required effective communication, coordination, and understanding of theoretical concepts to translate them into a working application. Each team member gained practical experience in handling a multi-faceted project, integrating backend and frontend components, and ensuring system functionality. The knowledge acquired through this project will be beneficial for future coursework and professional work in areas of database management, web app development, and system architecture.

However, during the website development, we have encountered many issues with our project structure where features do not work as intended; but we will be able to get it fixed and unimplemented features will also be added.

3.3 References

<https://www.w3schools.com/>

<https://erdplus.com/>

<https://github.com/HeomapTV123/Recipes.github.io>